Chapter 1
Matrices and Systems of Equations

Q1: Does a system of linear equations have a solution?

Q2: If it has, how many solutions?

1.1 Systems of Linear Equations

Q3: How to express the solution set?

linear system.

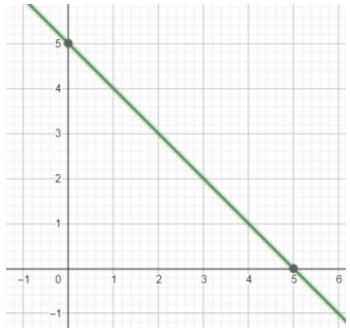
Objectives and key concepts

- Linear equations & Linear systems
  - Ax=b
  - Solution cases: consistent vs inconsistent
- Representing linear systems using matrices and vectors
  - For solutions **x** and right-hand side **b**: n-vector, n-space
  - For coefficients A: matrix
  - For linear system: augmented matrix [A|b]
- How to solve $n \times n$ linear systems
  - Elementary row operations on augmented matrices
  - Strictly triangular form: back-substitution

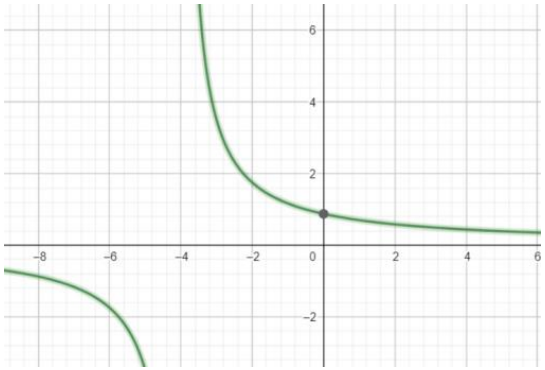
Linear Equations & Systems

$3x+2y+3z=10 \rightarrow$ plane.

- Linear equation: $x + y = 5$



- Non-linear equation:
  - $4x_1^2 + 6x_2 = 3$
  - $2x_1x_2 + 8x_2 = 7$
  - $5x_2^{1/2} = 9$

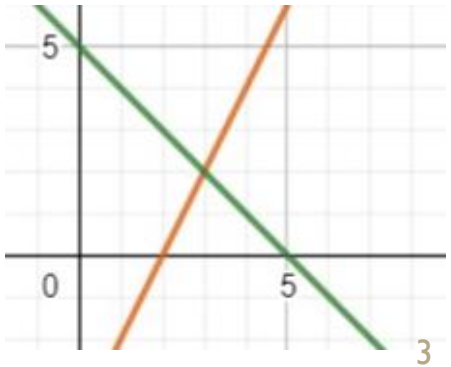


- Linear system:

2 x 2 → 2 variables.

2 equations

$$x + y = 5$$
$$2x - y = 4$$



nonlinear operations between variables.

variable multiplication, power, exponential

x^n 2^x, e^x

logrithmic square root ....

$\log_2 x$ $\sqrt{x}$.

Linear equations

scalar multiplication & addition

Definition/定义

A ***linear equation（线性方程）*** in n unknowns(未知数) $x_1, x_2, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where the coefficients（系数） $a_1, a_2, \dots, a_n$ and b are real numbers（实数）.

$a_1, a_2, \cdots, a_n, b \in R$

real number space.

$3x + 5y = 10$

$\sqrt{6}x + 2^5y = e^2$ ✓

Linear systems/线性方程组

Definition

A ***linear system*** of m equations in n unknowns $x_1, x_2, \ldots, x_n$ is a system of the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \qquad\qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

where the coefficients a_{ij}'s and b_j's are all real numbers.

The system is an $m \times n$ linear system.

Goal: to solve linear systems.

Question: What is a solution to a linear system? How many solutions are there?

Solution of linear systems

- Definition:
  A ***solution*** of an $m \times n$ system is an ordered *n-tuple of numbers*(n 元组) $x_1, x_2, \ldots, x_n$ that satisfies all the equations of the system.
  - An *inconsistent(不相容)* linear system has **no** solution;
  - A *consistent* *(相容)* system has **one** solution or **many** solutions.
- ***The solution set*** – the set of all solutions of a linear system.
- ***Equivalent systems*** – two linear systems with the same solution set.

1. Solution.

$(s_1, s_2, \cdots, s_n)$ is a solution to an $m \times n$ linear system. If let $x_1 = s_1$, $x_2 = s_2$, $\cdots$, $x_n = s_n$, and substitute them back to the m equations. the left-hand side of all the equations should be equal to the right-hand side.

Eg. ① $\begin{cases} x+y=5 \\ x-y=1 \end{cases}$ $(3,2)$ is a solution to this system.

Sine $x=3$ $y=2$.

$\begin{cases} 3+2=5 \\ 3-2=1 \end{cases}$

Solution set: $\{(3,2)\}$

② $\begin{cases} x+y=5 \\ 2x+2y=10 \end{cases}$ 2×2.

$(3,2)$ is a solution

$(1,4)$ is a solution

$(-1,6)$ is a solution

$\downarrow$

has infinitely many solutions.

Solution set: $\{(3,2), (1,4), (-1,6), \cdots\}$

③ No solution: Solution set: emp set. ϕ.

2. $\forall$: any / every / for all.

3 cases of ✡ solutions

$\forall$ $m \times n$ linear system.

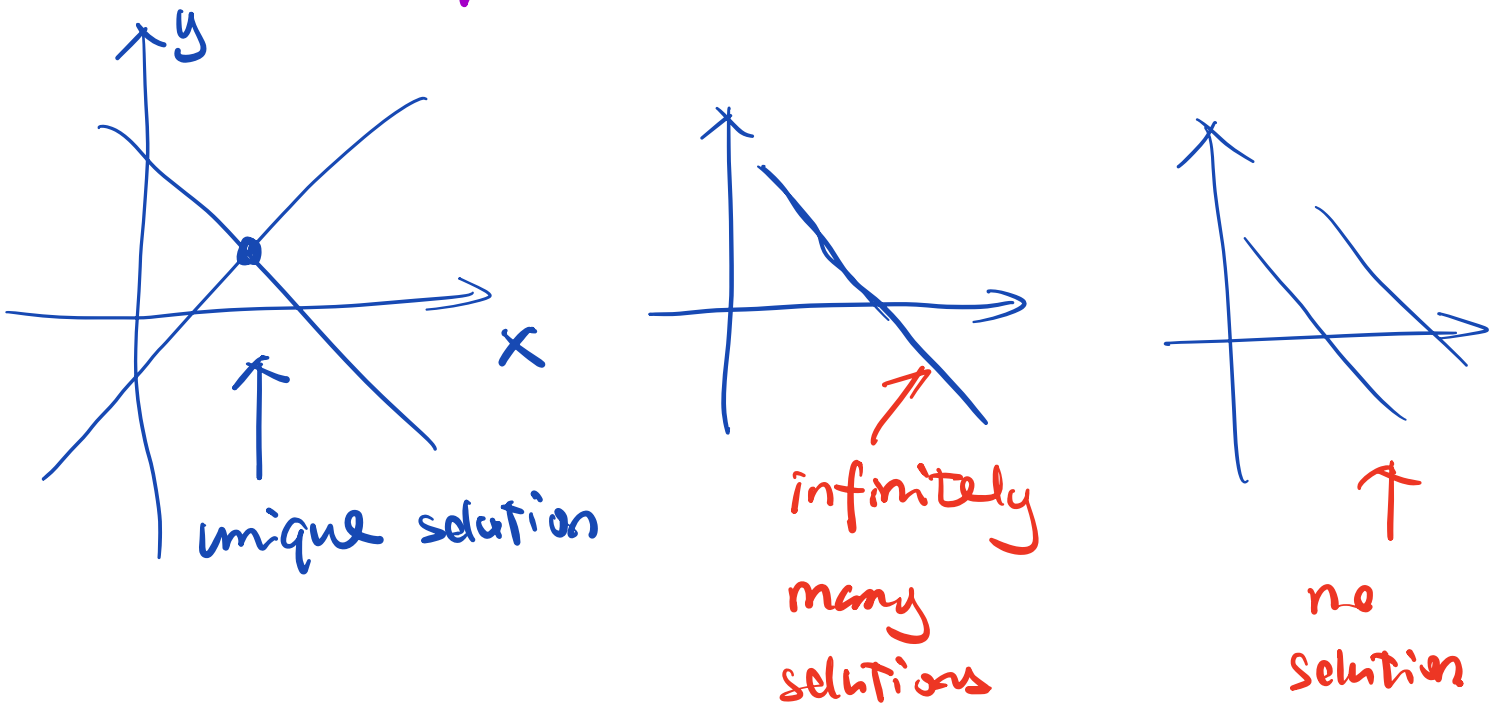
- consistent
  - unique
  - infinitely many
- inconsistent
  - no solutions

3. Q: Can we obtain exactly 2 solutions to a linear system?

No.

Eg. Consider 2×2 linear system.

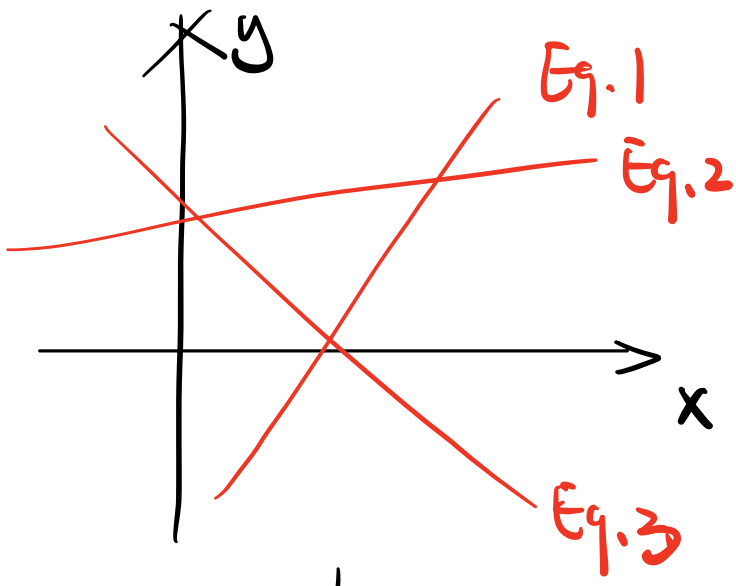
2 eqs. 2 variables.



4. If you have found 2 solutions to a linear system, it must have infinitely many solutions.

3×2
m×n

5. Question ① what is the size of this system?



② how many solutions?

No solution
↓
inconsistent

$$\begin{cases} x+y=2 \\ x-y=0 \\ 2x+2y=1 \end{cases} \Rightarrow \begin{cases} x=1 \\ y=1 \end{cases}$$

$2+2=4 \neq 1$

6. set A = set B

$\{1,3\} = \{1,3\} = \{3,1\}$

Eg. LS1 $\begin{cases} x+y+z=1 \\ y=z \end{cases}$ LS2 $\begin{cases} 2x+y+z=2 \\ y=z \end{cases}$

Equivalent? No.

$(1,0,0)$ is a solution $(1,0,0)$ is a solution

$(-1,1,1)$ is a solution $-2+1+1=-2 \neq 2$

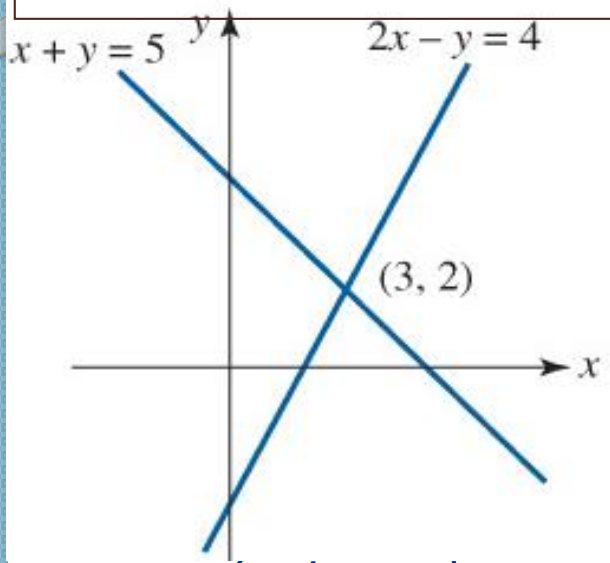
not a solution

$2x+2y=5 \iff 6x+6y=15$

1×2

Solution cases (2x2) 定解情况

One linear equation in 2 variables corresponds to a line in the two-dimensional (2D) plane.



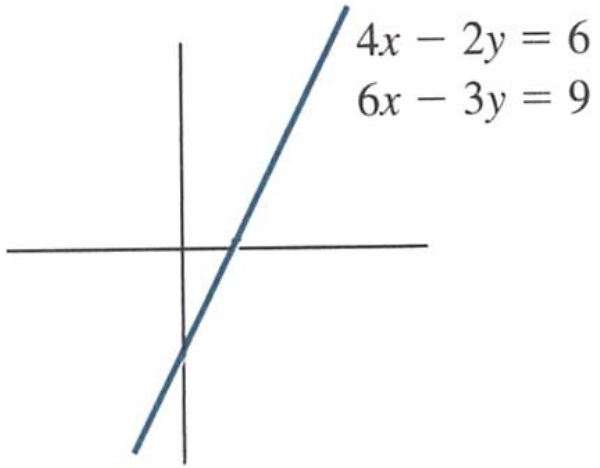
Unique (only one) solution

$$x + y = 5$$
$$2x - y = 4$$

Lines intersect at (3, 2)

Unique solution:

$$x = 3, y = 2.$$

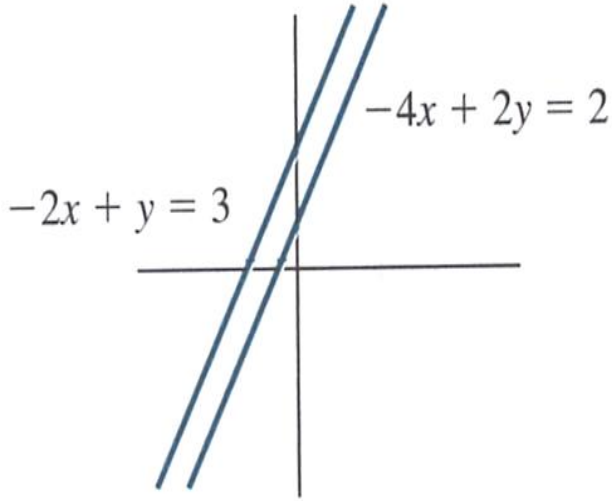


Many solutions

$$4x - 2y = 6$$
$$6x - 3y = 9$$

Both equations have the same graph. Any point on the graph is a solution.

Many solutions.



No solution

$$-2x + y = 3$$
$$-4x + 2y = 2$$

Lines are parallel.

No point of intersection

No solution.

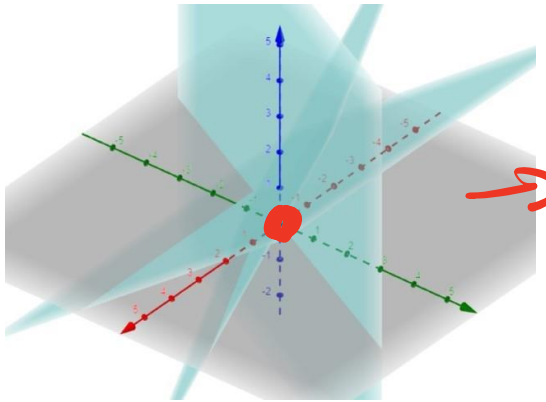
consistent (相容)

Inconsistent（不相容）

Solution cases (3x3)

A linear equation in three variables corresponds to a plane in three-dimensional (3D) space.

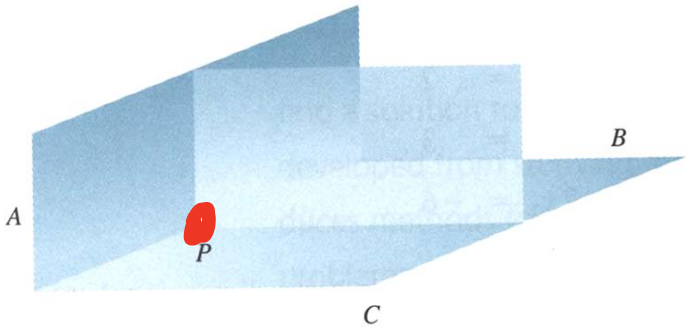
$$\begin{aligned} 2x - y &= 0 \\ -x + 2y - z &= -1 \\ -3y + 4z &= 4 \end{aligned}$$

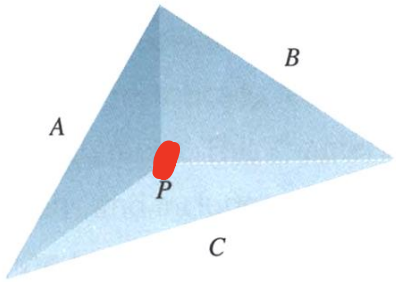


→ unique

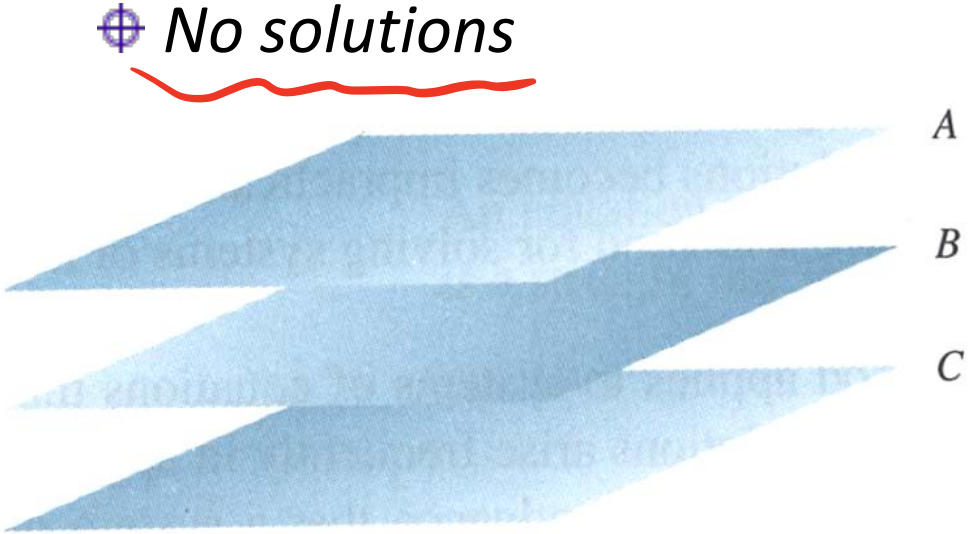
※ Systems of three linear equations in three variables:

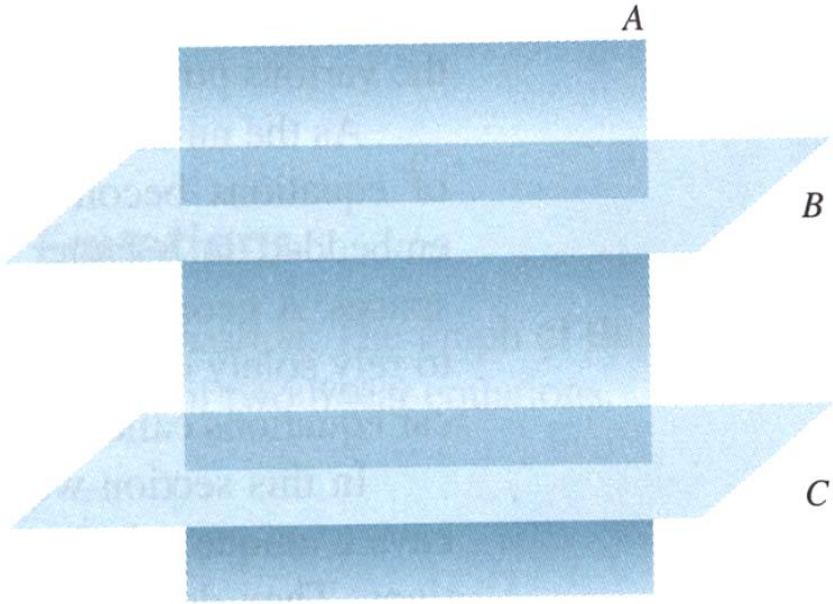
- *Unique solution*



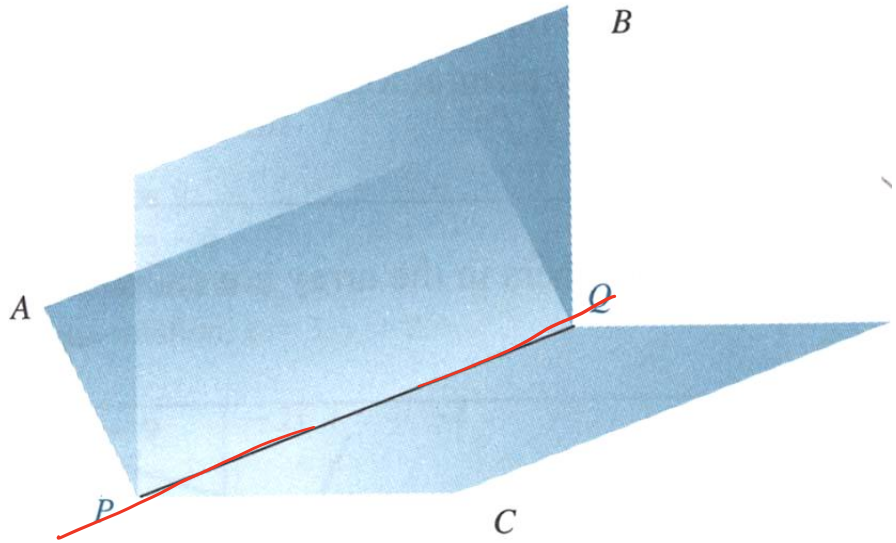


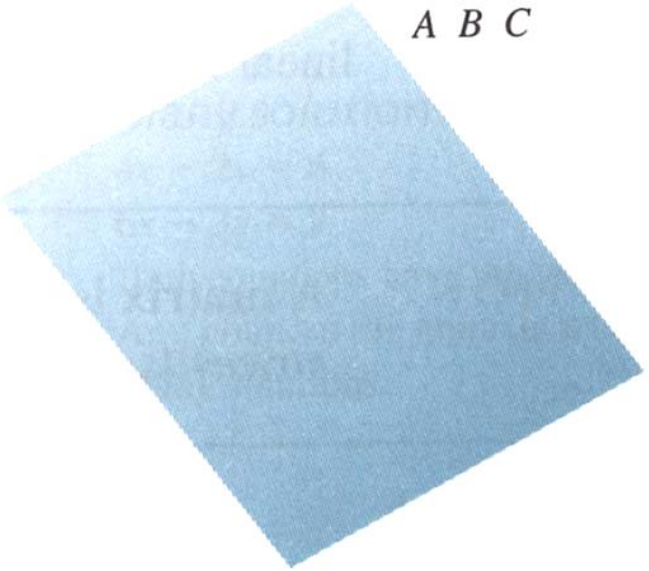
- *No solutions*





- *Many solutions*





Extend to 4D? And more?



$$\begin{cases} 3x_1 + 2x_2 + 5x_3 + 4x_4 = 10 \\ 3x_1 - 2x_5 = 9 \end{cases}$$

n-dimensional real space = Euclidean n-space.

- Let $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$ be a tuple of n real numbers.
  - For example, geometrically(几何上讲), $\mathbb{R}^2$ is a set of all points (x_1, x_2) in the plane.
- The set of all such tuples is denoted as $\mathbb{R}^n$, called real coordinate space of dimension n.
- In linear algebra, we commonly call $\boldsymbol{x}$ in $\mathbb{R}^n$ a vector(向量), and write it vertically as a column vector:

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

- The right-hand side constants in a linear system also forms a vector:

$$\boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \quad \in \mathbb{R}^{m \times 1}$$

$R = R^1$ real number space.

$R^n = R^{n\times 1}$ $\forall$: every / any / for all

$\forall x \in R^n$ → real

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \qquad x_i \in R \quad i = 1, 2, \cdots, n$$

column vector.

Eg. $\begin{bmatrix} 3 \\ 2 \end{bmatrix} \in R^2 = R^{2\times 1}$

$\neq$

$[3 \;\; 2]$

$\vec{y} = [y_1 \; y_2 \cdots y_n] \in R^{1\times n}$

row vector.

$R^n = R^{n\times 1}$

$\neq$

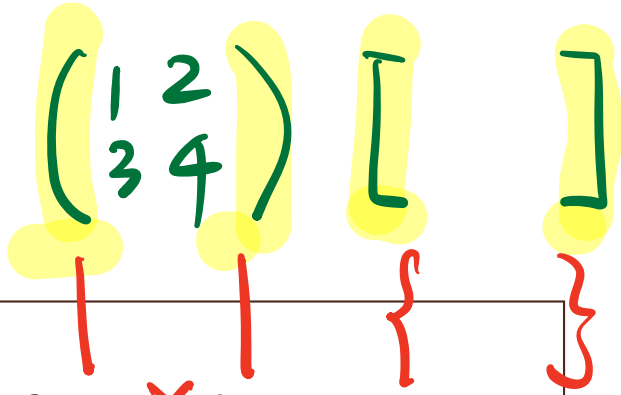
$R^{1\times n}$

a_{ij} : is the entry in the i-th row and j-th column.

rows ↑ m × n ↑ columns

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Matrix（矩阵）



Definition 矩阵　　矩形数组

- A ***matrix*** is a rectangular array of numbers.
- The numbers in the array are called the ***entries*** of the matrix.

元素

A: n×n

$$A=\begin{bmatrix} 2 & 3 & -4 \\ 7 & 5 & -1 \end{bmatrix} \quad B=\begin{bmatrix} 7 & 1 \\ 0 & 5 \\ -8 & 3 \end{bmatrix} \quad C=\begin{bmatrix} 3 & 5 & 6 \\ 0 & -2 & 5 \\ 8 & 9 & 12 \end{bmatrix}$$

2×3　　3×2　　3×3 → square matrix

Note there are square brackets[方括号] in left and right sides of the matrix.

Matrix/矩阵, and Vector/向量

Matrices with only one row（行） or one column（列） are called n-vectors

- A $1 \times n$ matrix is a *row vector*
- An $n \times 1$ matrix is a *column vector;* $\mathbb{R}^n$

$$\begin{bmatrix} 1 & 0 & 3 \\ -2 & 4 & 5 \end{bmatrix}$$

2×3 matrix

$$\begin{bmatrix} 2 & 5 & 7 \\ -9 & 0 & 1 \\ -3 & 5 & 8 \end{bmatrix}$$

3×3 matrix

square matrix

$$\begin{bmatrix} -2 & 4 & 5 \end{bmatrix} \in \mathbb{R}^{1\times 3}$$

1×3 matrix

row vector

$$\begin{bmatrix} 7 \\ 1 \\ 8 \end{bmatrix} \in \mathbb{R}^3$$

3×1 matrix

column vector

① For $\mathbb{R}$, $\forall x \in \mathbb{R}$

$x \cdot 1 = 1 \cdot x = x$

$\forall x \neq 0 \in \mathbb{R}$

$x \cdot x^{-1} = x^{-1} \cdot x = 1$

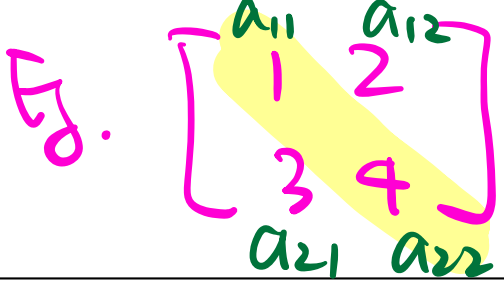
Identity matrix.

I.

① $\forall$ $m \times n$ A: $A \times I = A$

② If A^{-1} exists. $A \times A^{-1} = I$

Matrix, Vector



Identity Matrices （单位矩阵） 对角元素 n×n

A square matrix with diagonal value equals to 1 and other entries equal to 0, denoted as I_n,where the subscript (n) represent the size.

下标

Example

main diagonal: $\{a_{ii}, i=1,\cdots,n\}$

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$$

Question: $I_4 = ?$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4\times 4}$$

Row and column of a matrix

$$A = \begin{bmatrix} 2 & 3 & -4 \\ 7 & 5 & -1 \end{bmatrix}$$

2×3

$\begin{bmatrix} 2 & 3 & -4 \end{bmatrix}$ row 1

$\begin{bmatrix} 7 & 5 & -1 \end{bmatrix}$ row 2

$\begin{bmatrix} 2 \\ 7 \end{bmatrix}$ column 1

$\begin{bmatrix} 3 \\ 5 \end{bmatrix}$ column 2

$\begin{bmatrix} -4 \\ -1 \end{bmatrix}$ column 3

Matrices of a linear system (using an example)

Given the linear system

$$x_1 + x_2 + x_3 = 2$$
$$2x_1 + 3x_2 + x_3 = 3$$
$$x_1 - x_2 - 2x_3 = -6,$$

3×3

Organize all the coefficients in a 3x3 array of numbers:

Coefficient matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & -1 & -2 \end{bmatrix}$$

Eq.1 Eq.2 Eq.3

3×3

If we also write down the 3 right-hand side numbers:

Augmented matrix

增广矩阵

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 1 & -1 & -2 & -6 \end{array}\right]$$

3×4

How to solve a system of linear equations?

- Strategy: replace one system with an **equivalent (等价的) system** that is easier to solve. (2x2 example)
- Consider the following two systems:

I:

$$x_1 - 2x_2 = -1$$
$$-x_1 + 3x_2 = 3$$

(2)+(1)

II:

$$x_1 - 2x_2 = -1$$
$$x_2 = 2$$

$x_1 = -1 + 2x_2 = 3$

- System II. is easier to solve because it is clear from the second equation that $x_2 = 2$. Use this value in the first equation, we get $x_1 = 3$.
- System *I.* has the same solution as system *II.*, x=(3, 2).

How to solve a system of linear equations?

- Three operations that can be used on a system to obtain an equivalent system:
  - Type I: Interchange the order of two equations:

$$\begin{matrix} x+y=5 \\ 2x-y=4 \end{matrix} \sim \begin{matrix} 2x-y=4 \\ x+y=5 \end{matrix}$$

$R_i \Leftrightarrow R_j$

  - Type II: Multiply (divide) the same nonzero real number on both sides of equations:

Scaling αR_i $\alpha \neq 0$ $\frac{1}{3}x+\frac{1}{3}y=\frac{2}{3} \sim x+y=2$

  - Type III: Add to (subtract from) one equation a multiple of another equation:

Replacement. $R_j + \alpha R_i$

$$\begin{matrix} x+y=5 \\ x-y=3 \end{matrix} \overset{R_2+R_1}{\sim} \begin{matrix} x+y=5 \\ 2x=8 \end{matrix}$$

- Strategy:
  - Obtain an equivalent strictly triangular form.

$R_j + \alpha R_i$

destination row

Use

$x_1 + x_2 = 2$

$2x_1 - 2x_2 = 0$

$R_2 - 2R_1$

$-2R_2 + 3R_1$

$-2R_2$

$R_2 + 3R_1$

How to solve a system of linear equations?

Keep this example in mind:

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 4 \\ 0x_1 + 2x_2 + 5x_3 &= 4 \\ 0x_1 + 0x_2 + 3x_3 &= 6 \end{aligned}$$

An $n \times n$ system is said to be in strictly triangular form if in the kth equation, the coefficients of the first $k-1$ variables are all zero, and the coefficient of x_k is nonzero.

A linear system in strictly triangular form has coefficient matrix of the following form:

$$\begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{bmatrix}$$

① $n \times n$

② $a_{ki} = 0 \quad i = 1, 2, \cdots, k-1$

$\Updownarrow$

$a_{k1}=0$ & $a_{k2}=0$ & $\cdots$ & $a_{k,k-1}=0$

③ $a_{kk} \neq 0 \quad k = 1, \cdots, n$

Q1: $x_1 + 2x_2 + 3x_3 = 4$

$2x_2 = 5$

$-5x_3 = 6$

✓ Is it in S.T.F?

① 3×3

② ✓ ③ ✓

Q2: $x_1 + 2x_2 + 3x_3 = 4$

$0x_2 + x_3 = 5$

$-5x_3 = 6$

is not in S.T.F.

Type III $\quad R_j + \alpha R_i$

$-R_1 + 3R_2$ ✗ → $-R_1$ $R_1 + 3R_2$

$R_2 - 2R_1$

$R_3 + 3R_1$

$\longrightarrow$

$2R_2$

$R_2 - 2R_1$

$\longrightarrow$

$\times$

$\downarrow$

$2R_2$ $\longrightarrow$ $R_2 - 2R_1$ $\longrightarrow$

Example

Linear System:

$$3x_1 + 2x_2 - x_3 = -2$$
$$-3x_1 - x_2 + x_3 = 5$$
$$3x_1 + 2x_2 + x_3 = 2$$

Add the first equation to the second equation:

(R2 becomes R2 + R1)

$$3x_1 + 2x_2 - x_3 = -2$$
$$x_2 = 3$$
$$3x_1 + 2x_2 + x_3 = 2$$

Subtract first equation from the third equation:

(R3 becomes R3 – R1)

$$3x_1 + 2x_2 - x_3 = -2$$
$$x_2 = 3$$
$$2x_3 = 4$$

Then, use *back substitution* to get the solution:

- From equation 3: $x_3 = 2$
- From equation 2: $x_2 = 3$
- From equation 1: $x_1 = \frac{-2-2x_2+x_3}{3} = -2$

$(-2, 3, 2)$ is a solution unique to this linear system

From operations on linear systems to row operations on matrices

Linear systems correspond to augmented matrices.

For example:

$$x_1 + x_2 + x_3 = 2$$
$$2x_1 + 3x_2 + x_3 = 3$$
$$x_1 - x_2 - 2x_3 = -6$$

3x3

is the same as

coefficient matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 1 & -1 & -2 & -6 \end{array}\right]$$

3x4

Therefore, 3 operations used to obtain equivalent systems, correspond to 3 row operations on the augmented matrix.

$n \times n$ linear system $\longrightarrow$ augmented matrix

↓ Type I, II, III

Equivalent system

in S.T.F

↓ Type I, II, III

row operations

row equivalent

matrix

From operations on linear systems to row operations on matrices

Elementary Row Operations: 行初等变换

- Interchange two rows.
- Multiply a row by a nonzero real number.
- Replace a row by its sum with a multiple of another row.

- **Row equivalent matrices:** Two matrices where one matrix can be transformed into the other matrix by a sequence of elementary row operations.
- **Fact about Row Equivalence:** If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

Notation

$\overset{R1+(-3)R2}{\sim}$

Add (-3)× R2 to R1, hence R1 is changed but R2 keep intact.

For example:

(1) $\left[\begin{array}{ccc|c} 1 & -2 & 4 & 12 \\ 2 & -1 & 5 & 18 \\ -1 & 3 & -3 & -8 \end{array}\right] \overset{R1+2R2}{\sim} \left[\begin{array}{ccc|c} 5 & -4 & 14 & 48 \\ 2 & -1 & 5 & 18 \\ -1 & 3 & -3 & -8 \end{array}\right]$

(2) $\left[\begin{array}{ccc|c} 1 & -2 & 4 & 12 \\ 2 & -1 & 5 & 18 \\ -1 & 3 & -3 & -8 \end{array}\right] \overset{R2+2R1}{\sim} \left[\begin{array}{ccc|c} 1 & -2 & 4 & 12 \\ 4 & -5 & 13 & 42 \\ -1 & 3 & -3 & -8 \end{array}\right]$

Example

Solving the following system of linear equation

$$x_1 + x_2 + x_3 = 2$$
$$2x_1 + 3x_2 + x_3 = 3$$
$$x_1 - x_2 - 2x_3 = -6$$

3x3

Solution

Pivot: The first non-zero entry in the pivotal row

Equation Method

Initial system:

$$x_1 + x_2 + x_3 = 2$$
$$2x_1 + 3x_2 + x_3 = 3$$
$$x_1 - x_2 - 2x_3 = -6$$

Matrix Method

Augmented matrix:

pivot

Pivotal row

$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 1 & -1 & -2 & -6 \end{bmatrix}$$

$$x_1 + x_2 + x_3 = 2$$

Eq2+(−2)Eq1 $\quad x_2 - x_3 = -1$

Eq3+(−1)Eq1 $\quad -2x_2 - 3x_3 = -8$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & -3 & -8 \end{bmatrix}$$

R2+(−2)R1

R3+(−1)R1

$$x_1 + x_2 + x_3 = 2$$
$$x_2 - x_3 = -1$$
$$-2x_2 - 3x_3 = -8$$

Pivot row

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & -3 & -8 \end{array}\right]$$

pivot

R_3+2R_2

$$x_1 + x_2 + x_3 = 2$$
$$x_2 - x_3 = -1$$
$$-5x_3 = -10$$

Eq3+(2)Eq2

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -5 & -10 \end{array}\right]$$

R3+(2)R2

$x_3 = 2$

$x_2 = -1 + x_3 = 1$

$x_1 = -1$

The solution is

$$x_1 = -1, x_2 = 1, x_3 = 2$$

The solution is

$$x_1 = -1, x_2 = 1, x_3 = 2$$

$(-1, 1, 2)$ is a solution to this LS.

Check! Is this a solution of the original system?

(1). For an $n \times n$ linear system.

(a) It can be simplified to the S.T.F

if and only if

(b) the linear system has the unique solution

(a) → (b)

S.T.F

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1,n-1}x_{n-1} + a_{1n}x_n = b_1$$

$$a_{22}x_2 + \cdots + a_{2,n-1}x_{n-1} + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{n-1,n-1}x_{n-1} + a_{n-1,n}x_n = b_{n-1}$$

$$a_{nn}x_n = b_n$$

$a_{11} \neq 0$ & $a_{22} \neq 0$ & $\cdots$ & $a_{nn} \neq 0$

$x_1 = \frac{\quad}{a_{11} \neq 0}$ unique

$x_2 = \frac{\quad}{a_{22} \neq 0}$ unique

$x_{n-1} = \frac{b_{n-1} - a_{n-1,n}x_n}{a_{n-1,n-1} \neq 0}$ unique.

$x_n = \frac{b_n}{a_{nn}}$ unique

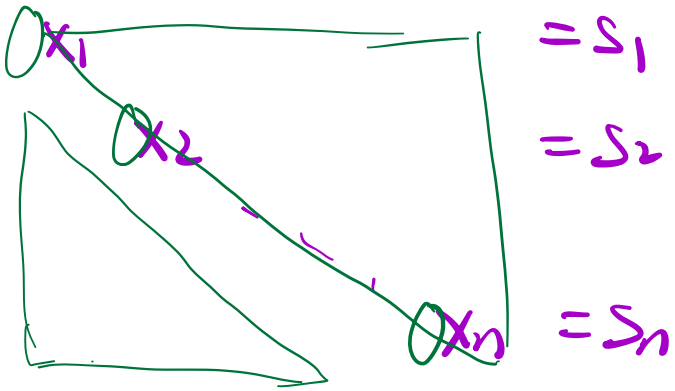
This $n \times n$ linear system has a unique solution.

(b) → (a)

Suppose that $(s_1, s_2, \cdots, s_n)$ is

the unique solution to the LS.

$$\begin{cases} x_1 = s_1 \\ x_2 = s_2 \\ \vdots \\ x_n = s_n \end{cases}$$



is S.T.F

Objectives and key concepts

- Linear equations & Linear systems
  - Solution cases: consistent vs inconsistent
- Representing linear systems using matrices and vectors
  - For solutions **x** and right-hand side **b**: n-vector, n-space
  - For coefficients A: matrix
  - For linear system: augmented matrix [A|b]
- How to solve $n \times n$ linear systems
  - Equivalent systems: 3 operations
  - Strick triangular form, back-substitution
  - Elementary row operations on augmented matrices